Session 3: Orbit Transfers and Delta-V

Task 1: Explain why both Hohmann burns are purely tangential, and why a plane change is cheapest at apoapsis.

Tangential Hohmann Burns

- Hohmann transfer burns are applied strictly parallel/ tangential to the velocity vector $\Delta \boldsymbol{v} || \boldsymbol{v}$. This maximizes instantaneous mechanical energy gain per unit impulse according to $\frac{dE}{dt} = \boldsymbol{v} \cdot \boldsymbol{a}$.
- Delivering Δv at maximum kinetic energy v leverages the **Oberth effect**, maximizing a_{trans} gain during the periapsis burn and orbital circularization during the apoapsis burn.

Plane changes at apoapsis:

- The Δv cost for a pure inclination change Δi is given by $\Delta v_{plane} = 2v \sin\left(\frac{\Delta i}{2}\right)$
- Because $v = \sqrt{\mu\left(\frac{2}{r} - \frac{1}{a}\right)}$ is at its absolute minimum at apoapsis $\left(r_{apogee}\right)$, performing plane alignment at apoapsis drastically reduces propellant mass requirements relative to periapsis maneuvers.

Task 2: Hohmann Transfer

For Earth orbital calculations, $\mu = 398600\ km^3/s^2$, and $R_E = 6378\ km$.

1. **LEO** $(500\ km)$ **to GEO transfer**
   - **Initial & Final Radii:** $r_1 = 6378 + 500 = 6878\ km, r_2 = 6378 + 35786 = 42164\ km$
   - **Orbital velocities:** $v_1 = \sqrt{\frac{\mu}{r_1}} = \sqrt{\frac{398600}{6878}} = 7.613\,\frac{km}{s}, v_2 = \sqrt{\frac{\mu}{r_2}} = \sqrt{\frac{398600}{42164}} = 3.075\,\frac{km}{s}$
   - **Transfer Semi-major Axis:** $a_{trans} = \frac{r_1 + r_2}{\mathbf{2}} = \frac{6878+42164}{2} = 24521.5\ km$
   - **Transfer Velocities:**
     - $v_{t1} = \sqrt{\mu\left(\frac{2}{r_1} - \frac{1}{a_{trans}}\right)} = \sqrt{398600 \cdot \left(\frac{2}{6878} - \frac{1}{24521.5}\right)} = 9.983\ km/s$

- $v_{t2} = \sqrt{\mu\left(\frac{2}{r_2} - \frac{1}{a_{trans}}\right)} = \sqrt{398600 \cdot \left(\frac{2}{42164} - \frac{1}{24521.5}\right)} = 1.628\ km/s$

- **Impulses & Time:**
  - $\boldsymbol{\Delta v_1} = v_{t1} - v_1 = 9.983 - 7.613 = 2.37\ km/s$
  - $\mathbf{\Delta v_2} = v_2 - v_{t2} = 3.075 - 1.628 = 1.447\ km/s$
  - $\boldsymbol{\Delta v_{total}} = \Delta v_1 + \Delta v_2 = 2.37 + 1.447 = 3.817\ km/s$
  - **Transfer duration:** $t_{trans} = \pi\sqrt{\frac{a_{trans}^3}{\mu}} = \pi\sqrt{\frac{24521.5^3}{398600}} = 19107.409\ s \approx 5.3\ h$

2. LEO $(400\ km)$ **to MEO** $(20200\ km)$

- **Initial & Final Radii:** $r_1 = 6378 + 400 = 6778\ km, r_2 = 6378 + 20200 = 26578\ km$
- **Orbital velocities:** $v_1 = \sqrt{\frac{\mu}{r_1}} = \sqrt{\frac{398600}{6778}} = 7.669\frac{km}{s}, v_2 = \sqrt{\frac{\mu}{r_2}} = \sqrt{\frac{398600}{26578}} = 3.873\frac{km}{s}$
- **Transfer Semi-major Axis:** $a_{trans} = \frac{r_1 + r_2}{2} = \frac{6778 + 26578}{2} = 16678\ km$
- **Transfer Velocities:**
  - $v_{t1} = \sqrt{\mu\left(\frac{2}{r_1} - \frac{1}{a_{trans}}\right)} = \sqrt{398600 \cdot \left(\frac{2}{6778} - \frac{1}{16678}\right)} = 9.681\ km/s$
  - $v_{t2} = \sqrt{\mu\left(\frac{2}{r_2} - \frac{1}{a_{trans}}\right)} = \sqrt{398600 \cdot \left(\frac{2}{26578} - \frac{1}{16678}\right)} = 2.469\ km/s$
- **Impulses & Time:**
  - $\boldsymbol{\Delta v_1} = v_{t1} - v_1 = 9.681 - 7.669 = 2.012\ km/s$
  - $\mathbf{\Delta v_2} = v_2 - v_{t2} = 3.872 - 2.469 = 1.403\ km/s$
  - $\boldsymbol{\Delta v_{total}} = \Delta v_1 + \Delta v_2 = 2.012 + 1.403 = 3.415\ km/s$
  - **Transfer duration:** $t_{trans} = \pi\sqrt{\frac{a_{trans}^3}{\mu}} = \pi\sqrt{\frac{16678^3}{398600}} = 10717.591\ s$

Task 3: Bi-Elliptic vs Hohmann Comparision

Setting $R = \frac{r_2}{r_1} = 20, r_b = 40r_1$

- **Hohmann Transfer Total** Δv:
  - Transfer semi-major axis $a_h = \frac{r_1 + r_2}{2} = \frac{r_1 + 20r_1}{2} = 10.5r_1$

- $\Delta v_1 = v_{t1} - v_1 = \sqrt{\mu\left(\frac{2}{r_1} - \frac{1}{a_h}\right)} - \sqrt{\frac{\mu}{r_1}} = \sqrt{\mu}\left(\sqrt{\frac{2}{r_1} - \frac{1}{10.5r_1}} - \sqrt{\frac{1}{r_1}}\right) = \sqrt{\mu}\sqrt{\frac{1}{r_1}}\left(\sqrt{\frac{40}{21}} - 1\right) = 0.38v_1$
- $\Delta v_2 = v_2 - v_{t2} = \sqrt{\frac{\mu}{r_2}} - \sqrt{\mu\left(\frac{2}{r_2} - \frac{1}{a_h}\right)} = \sqrt{\mu}\left(\sqrt{\frac{1}{20r_1}} - \sqrt{\frac{2}{20r_1} - \frac{1}{10.5r_1}}\right) = 0.15v_1$
- $\Delta v_h = \Delta v_1 + \Delta v_2 = 0.38v_1 + 0.15v_1 = 0.53v_1$
- Applied to LEO $(500\ km)$: $\Delta v_{Hohmann} = 0.53v_1 = 0.53\sqrt{\frac{\mu}{r_1}} = 0.53\sqrt{\frac{398600}{6378+500}} = 4.03\ km/s$

- **Bi-Elliptic Transfer Total**:
  - Ellipse 1 $(r_1 \rightarrow r_b)$:

$$a_1 = \frac{r_1 + r_b}{2} = \frac{r_1 + 40r_1}{2} = 20.5r_b$$

$$\Delta v_1 = v_{t1} - v_1 = \sqrt{\mu\left(\frac{2}{r_1} - \frac{1}{a_1}\right)} - \sqrt{\frac{\mu}{r_1}} = \sqrt{\mu}\left(\sqrt{\frac{2}{r_1} - \frac{1}{20.5r_1}} - \sqrt{\frac{1}{r_1}}\right)$$

$$= \sqrt{\mu}\sqrt{\frac{1}{r_1}}\left(\sqrt{\frac{80}{41}} - 1\right) = 0.40v_1$$

  - Ellipse 2 $(r_b \rightarrow r_2)$:

$$a_2 = \frac{r_b + r_2}{2} = \frac{40r_1 + 20r_1}{2} = 30r_1$$

$$\Delta v_2 = \sqrt{\mu\left(\frac{2}{r_b} - \frac{1}{a_2}\right)} - \sqrt{\mu\left(\frac{2}{r_b} - \frac{1}{a_1}\right)}$$

$$= \sqrt{\mu}\left(\sqrt{\frac{2}{40r_1} - \frac{1}{30r_1}} - \sqrt{\frac{2}{40r_1} - \frac{1}{20.5r_1}}\right)$$

$$= \sqrt{\mu}\sqrt{\frac{1}{r_1}}\left(\sqrt{\frac{1}{60}} - \sqrt{\frac{1}{820}}\right) = 0.09v_1$$

- Final burn (r_2):

$$\Delta v_3 = \sqrt{\frac{\mu}{r_2}} - \sqrt{\mu\left(\frac{2}{r_2} - \frac{1}{a_2}\right)} = \sqrt{\mu}\left(\sqrt{\frac{1}{20r_1}} - \sqrt{\frac{2}{20}r_1 - \frac{1}{30}r_1}\right)$$

$$= \sqrt{\mu}\sqrt{\frac{1}{r_1}}\left(\sqrt{\frac{1}{20}} - \sqrt{\frac{1}{10} - \frac{1}{30}}\right) = -0.035v_1$$

- Applied to LEO ($500\,km$): $\Delta v_{Bi-Elliptic} = \Delta v_1 + |\Delta v_2| + \Delta v_3 = 0.4v_1 + 0.09v_1 - 0.035v_1 = 0.455\sqrt{\frac{\mu}{r_1}} = 0.455\sqrt{\frac{398600}{6378+500}} = 3.46\ km/s$

- **Trade-Off Analysis**: Bi-elliptic saves $\approx 0.57\ km/s$ of Δv, but incurs a **massive time penalty** due to extending apogee out to $40r_1$, increasing total transfer duration from hours to days.

Task 4: Combined burn

When launching from an inclined site (e.g., $i = 51.6^o$ or 28.5^o), combining orbital circularization (Δv_2) and inclination change (Δi) into a single apogee vector burn significantly reduces overall expenditure via the Law of Cosines:

$$\Delta v_{combined} = \sqrt{v_{trans,apo}^2 + v_{GEO}^2 - 2v_{trans,apo}^2 v_{GEO}\cos(\Delta i)}$$

- **Case 1: Launch from** $i = 51.6^o$
  - $v_{trans,apo} = 1.628\frac{km}{s}, v_{GEO} = 3.075\ km/s$
  - $\Delta v_{combined} = \sqrt{1.628^2 + 3.075^2 - 2 \cdot 1.628 \cdot 3.075 \cdot \cos(51.6^o)} = 2.43\ km/s$
  - Comparison: Performing separate burns requires $\Delta v_2 + \Delta v_{plane} = 1.447 + 2v_{GEO}\sin\left(\frac{\Delta i}{2}\right) = 1.447 + 2 \cdot \sin\left(\frac{51.6}{2}\right) = 4.12\ km/s.$ Combined burn saves $1.69\ km/s$
- **Case 2: Launch from** $i = 28.5^o$

- $\Delta v_{combined} = \sqrt{1.628^2 + 3.075^2 - 2 \cdot 1.628 \cdot 3.075 \cdot \cos(28.5^o)} = 1.82\ km/s$

Code for Task 2, 3, 4.

```
import numpy as np

# ==========================================
# CONSTANTS & PHYSICAL PARAMETERS
# ==========================================
MU = 398600   # Earth gravitational parameter (km^3/s^2)
R_E = 6378    # Earth equatorial radius (km)

def hohmann_transfer(r1, r2):
    """Calculates Hohmann transfer delta-v components and transfer duration."""
    v1 = np.sqrt(MU / r1)
    v2 = np.sqrt(MU / r2)
    a_trans = (r1 + r2) / 2.0

    v_trans_peri = np.sqrt(MU * (2.0 / r1 - 1.0 / a_trans))
    v_trans_apo = np.sqrt(MU * (2.0 / r2 - 1.0 / a_trans))

    dv1 = v_trans_peri - v1
    dv2 = v2 - v_trans_apo
    dv_total = dv1 + dv2
    t_trans_hrs = (np.pi * np.sqrt(a_trans**3 / MU)) / 3600.0

    return dv1, dv2, dv_total, t_trans_hrs, v_trans_apo, v2

# ==========================================
# TASK 2: HOHMANN TRANSFERS
# ==========================================
print("--- TASK 2: HOHMANN TRANSFERS ---")

# Task 2.1: LEO (500 km) to GEO
r1_500 = R_E + 500.0
r2_geo = 42164.14
dv1_geo, dv2_geo, dv_tot_geo, t_geo, v_trans_apo_geo, v_geo = hohmann_transfer(r1_500, r2_geo)

print(f"[LEO 500km -> GEO]")
print(f"  dv1: {dv1_geo:.2f} km/s | dv2: {dv2_geo:.2f} km/s | Total dv: {dv_tot_geo:.2f} km/s")
print(f"  Transfer Time: {t_geo:.2f} hours\n")

# Task 2.2: LEO (400 km) to MEO (20,200 km altitude)
r1_400 = R_E + 400.0
r2_meo = R_E + 20200.0
dv1_meo, dv2_meo, dv_tot_meo, t_meo, _, _ = hohmann_transfer(r1_400, r2_meo)

print(f"[LEO 400km -> MEO 20,200km]")
print(f"  dv1: {dv1_meo:.2f} km/s | dv2: {dv2_meo:.2f} km/s | Total dv: {dv_tot_meo:.2f} km/s")
print(f"  Transfer Time: {t_meo:.2f} hours\n")

# ==========================================
# TASK 3: BI-ELLIPTIC COMPARISON (R = 20, r_b = 40*r1)
# ==========================================
print("--- TASK 3: BI-ELLIPTIC VS HOHMANN (R = 20) ---")

r1_norm = 1.0
r2_norm = 20.0
rb_norm = 40.0
v1_phys = np.sqrt(MU / r1_500)

# Hohmann for R = 20
a_h = (r1_norm + r2_norm) / 2.0
v_tp1_h = np.sqrt(2.0/r1_norm - 1.0/a_h)
v_tp2_h = np.sqrt(2.0/r2_norm - 1.0/a_h)
v2_h = np.sqrt(1.0/r2_norm)
```

```
dv_h_tot = (v_tp1_h - 1.0) + (v2_h - v_tp2_h)
dv_h_phys = dv_h_tot * v1_phys

# Bi-elliptic for R = 20, r_b = 40
a1_b = (r1_norm + rb_norm) / 2.0
v_tp1_b = np.sqrt(2.0/r1_norm - 1.0/a1_b)
v_tpb1 = np.sqrt(2.0/rb_norm - 1.0/a1_b)

a2_b = (r2_norm + rb_norm) / 2.0
v_tpb2 = np.sqrt(2.0/rb_norm - 1.0/a2_b)
v_tp2_b = np.sqrt(2.0/r2_norm - 1.0/a2_b)

dv_bi_tot = (v_tp1_b - 1.0) + abs(v_tpb2 - v_tpb1) + (v2_h - v_tp2_b)
dv_bi_phys = dv_bi_tot * v1_phys

print(f"Hohmann Total (R=20):     {dv_h_phys:.2f} km/s")
print(f"Bi-elliptic Total (r_b=40): {dv_bi_phys:.2f} km/s")
print(f"Delta-v Savings:          {dv_h_phys - dv_bi_phys:.2f} km/s (Bi-elliptic wins)\n")

# ========================================
# TASK 4: COMBINED BURN (LEO 500km -> GEO with i = 51.6 deg)
# ========================================
print("--- TASK 4: COMBINED APOGEE BURN ---")

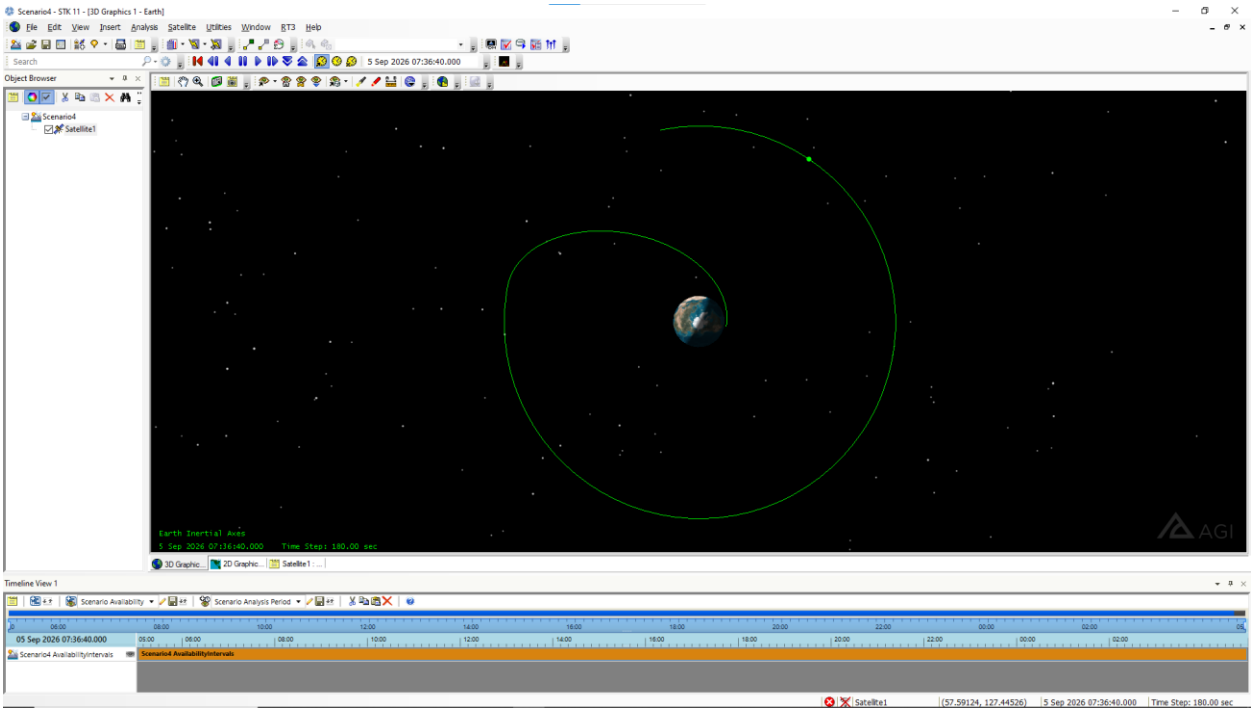
inc_deg = 51.6
inc_rad = np.radians(inc_deg)

# Combined Vector Burn via Law of Cosines
dv_combined = np.sqrt(v_trans_apo_geo**2 + v_geo**2 - 2.0 * v_trans_apo_geo * v_geo *
np.cos(inc_rad))

# Separate In-Plane Burn + Plane Change
dv_plane_only = 2.0 * v_geo * np.sin(inc_rad / 2.0)
dv_separate = dv2_geo + dv_plane_only

print(f"Combined Apogee Burn Cost: {dv_combined:.2f} km/s")
print(f"Separate Burns Cost:       {dv_separate:.2f} km/s")
print(f"Savings via Vector Combined Burn: {dv_separate - dv_combined:.2f} km/s")
```

Task 5: STK


Scenario4 - STK 11 - [3D Graphics 1 - Earth]
File Edit View Insert Analysis Satellite Utilities Window RT3 Help
Object Browser
Scenario4
Satellite1
5 Sep 2026 07:36:40.000
Earth Inertial Axes
5 Sep 2026 07:36:40.000 Time Step: 180.00 sec
AGI
3D Graphic...
2D Graphic...
Satellite1 ...

Timeline View 1
Scenario Availability
Scenario Analysis Period
05 Sep 2026 07:36:40.000
Scenario4 AvailabilityIntervals
Scenario4 AvailabilityIntervals
Satellite1
(57.59124, 127.44526)
5 Sep 2026 07:36:40.000
Time Step: 180.00 sec